Archit Mangrulkar

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Education

Indian Institute of Technology Kharagpur **Nov 2020 – May 2024**

Bachelor of Technology in Computer Science *CGPA 8.57/10.00*

- Earned a branch change into Computer Science from Mathematics, placing top 16 of 1800 on a 10.0 first-year CGPA

Work Experience

Samsung Research India (SRI-B) **Jul 2024 – Present**

Senior Software Engineer | Video Solutions, AI Computational Imaging Group *Bangalore, India*

- Own real-time C++ video features (**VDIS**, QuickTake, Hyperlapse) on Exynos, MediaTek and Qualcomm chipsets
- Ship stabilization as a native `.so` that plugs directly into the Android camera pipeline on every current flagship device
- Hold a hard **16.67 ms** budget per frame at 60 fps with 8K 10-bit HDR, tuning worst-case rather than mean latency
- Fuse asynchronous gyro (526 μs) and OIS (1 ms) streams onto the frame timeline, absorbing their timestamp jitter
- Cut per-frame compute with **ARM NEON SIMD** intrinsics, OpenCL kernels and cache-aware memory layout work
- Led commercialization for the S flagship and A/M/F series across the NA, EUR and Asian markets, clearing KPI gates
- Building video denoising and portrait-to-landscape video features now targeting the Flip, Fold and S flagship range
- Delivered 51 hours of C++ optimization and OpenCV training to **200+ engineers** across six sibling engineering teams

University of Alberta, Canada **May 2023 – Aug 2023**

Vision & Learning Lab | Multimodal Motion Capture *Edmonton, Canada*

- Built a large-scale multimodal dataset of humans in the wild, combining RGBD images, eye-tracking data and speech
- Drove a six **Azure Kinect** rig over sockets, holding every camera at a frame barrier before the capture could advance
- Wrote the **PyQt5** capture client with separate record and save threads so that the disk writes never stalled capture
- Recovered camera intrinsics with OpenCV chessboards after tracing bad registration to a 1536P versus 1080P mismatch
- Composed pairwise 4x4 transforms to derive RGB-to-depth extrinsics and land all six views in one reference frame
- Lifted **OpenPose** keypoints into 3D and fused duplicate detections across cameras with KMeans on joint distances

TU Dresden, Germany **May 2022 – Jul 2022**

Chair of Processor Design | Approximate Computing in Neural Networks *Dresden, Germany*

- Wrote neural operators as C static libraries for fast **8-bit FPGA** inference of quantized networks via TensorFlow Lite
- Explored cross-layer granularity with signed 8-bit lookup tables, polynomial regression and EvoApproxLib multipliers
- Cut inference time **2x** with **OpenMP** multithreading, autovectorization, compiler flag tuning and cache optimizations

Key Projects

Order Matching Engine | *C++17, lock-free, x86_64 & ARM64* | GitHub

- Built a C++17 limit order book and matching engine with LIMIT/IOC/FOK types and strict **price-time priority**
- Implemented FOK as a read-only walk of the opposite book, so an order that cannot fill ends up leaving zero state change
- Stored price levels as intrusive doubly-linked FIFOs with the list pointers living inside `Order`, so that cancels cost **O(1)**
- Eliminated `malloc` from the hot path with a 64B-aligned slab allocator and an intrusive free list of reusable orders
- Designed a Vyukov-style **lock-free SPMC ring** where consumers race a CAS'd head, claiming each order exactly once
- Measured p50 83 ns and **p99 291.7 ns** over 10M orders, timing each engine call with `RDTSC` on an untuned laptop
- Profiled `std::map` price-level lookup as the next O(log n) bottleneck and then scoped out a tick-indexed flat array rewrite
- Validated price-time ordering, IOC remainder cancels and FOK atomicity with a concurrent no-loss queue stress test

Lock-Free Concurrency Toolkit | *C++17, atomics, cache-aware* | GitHub

- Implemented a bounded **MPMC ring buffer** and a sharded hash map, both benchmarked against a mutex baseline
- Used per-slot sequence numbers over contiguous storage to stop torn reads and double-claims on any one single slot
- Killed false sharing with `alignas(64)` slot padding, plus an exponential backoff on every single failed CAS (`PAUSE`/`YIELD`)
- Measured the lock-free queue losing at **0.69x** to `std::mutex`, then root-caused and fixed the unfair harness behind it
- Sharded the map behind per-shard `shared_mutex`, a deliberate trade against hazard pointers or epoch reclamation
- Stress-tested **800K** ops over 4 producers and 4 consumers, plus 40K concurrent inserts verified across eight threads

Kalman Filters in Financial Markets | *state estimation, pairs trading* | GitHub

- Worked through the **Kalman Filter** as a recursive state estimator and wrote the theory up as a short white paper
- Applied it to pairs trading, letting the filter track the hedge ratio between two correlated names as a hidden state
- Benchmarked that adaptive hedge ratio against a rolling **OLS** regression to see where each approach breaks down
- Smoothed **MACD** by running several Kalman filters with different transition covariances across the same price signal
- Tuned the Kalman gain so that the noisiest measurements get down-weighted through every predict and correct cycle

ARMA-Based Trading Strategy | *time series, volatility modelling* | GitHub

- Checked the AAPL price series for stationarity with an **ADF** test before fitting any ARMA orders to daily returns
- Built an ARMA-driven strategy on AAPL that reached roughly **10%** maximum profit across the backtest window
- Raised that to **23%** by filtering entries with RSI, MACD and Golden Cross indicators built on both price and volume
- Modelled conditional volatility with an ARCH model and benchmarked AAPL's swings against the S&P 500 index
- Kept each strategy variant in a separate notebook so the contribution of every individual filter stayed easy to isolate

Technical Skills

Languages: C++ 17/20, C, Python, Bash, MATLAB, LaTeX; STL, templates, RAII, move semantics and constexpr
Concurrency: lock-free programming, C++ memory model and atomics, CAS with backoff, false sharing, cache padding
Performance: slab allocators, ARM NEON SIMD, OpenCL, OpenMP, multithreading, profiling, Android NDK and JNI
Tools: perf, gdb, CMake, Git, Docker, Linux/POSIX toolchain, ONNX Runtime, PyTorch, OpenCV, BackTrader, Jupyter

Patents

Geometrically Consistent Multi-Camera Stabilization for XR Headsets | *Patent Pending*

- Co-invented an AI-driven multi-camera stabilization method that enforces geometric consistency across the camera rig
- Solved AR world-anchor loss under rapid, dynamic head motion with reinforcement learning inside the stabilization loop
- Analyzed stabilization pipelines from **Apple, Meta** and Xreal to find architectural limits and shape our novel claims

Publications

AAAI Conference on Artificial Intelligence 2026 | *Under Review*

- Deployed a multi-branch video stability model in C++ and ONNX Runtime for a **15x speedup** over Python code
- Cut **model size by 72%** with static INT8 quantization, solving Android Binder IPC and GPU memory contention
- Replaced RAFT optical flow with NeuFlow v2, a 44x FLOPs reduction that finally made edge deployment feasible

COLING 2022 | *29th International Conference on Computational Linguistics* | ACL

- Co-authored and presented a shared-task paper on Stance and Premise Detection at the SMM4H workshop in 2022
- Achieved **0.636 F1** on the stance subtask using CovidTwitterBERT and 0.664 F1 on the premise task using BART-base
- Placed **first** on the workshop leaderboard using PoS, dependency-parse and Tf-Idf features with contrastive pretraining

Academic Achievements

- Secured **AIR 978** out of 150k candidates in JEE Advanced and **AIR 1517** out of 929k in JEE Mains, both in 2020
- MITACS Scholarship 2023 at the University of Alberta and Erasmus+ Scholarship 2022 at TU Dresden, Germany
- Won gold at DRDO's UGV Navigation Challenge (Inter IIT 10.0) and silver at DevRev's QA Challenge (11.0), 2022

Co-Curricular Activities

- **Data Head, Quant Club**: ran two Summer of Quant editions on derivatives and algo trading, drawing 5k+ footfall
- Member, Data Analytics Group: organized a national hackathon drawing 2100 team registrations and an ML workshop
- Student Welfare Group Mentor from 2021 to 2023, guiding first-year students through academic and personal transitions
- Won a bronze medal in General Championship Basketball 2022, representing Lala Lajpat Rai Hall at IIT Kharagpur